

The subcritical Métivier Schrödinger potential is essentially self-adjoint

Candidate solution packet for arXiv:1610.09850

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Status. Candidate full affirmative answer to Remark 4.8, likely valid, awaiting specialist review.

1 Source question

Bruno and Calzi ask in Remark 4.8 of [1], on PDF page 12, whether the Schrödinger operator associated with their weighted sub-Laplacian is essentially self-adjoint in the range $0 < \alpha < 2$.

Corollary 4.7. *For every $\alpha \geq 2$, $(\mathcal{L} + V_\alpha, C_c^\infty)$ is essentially self-adjoint and $\mathcal{L}^{w_\alpha}$ is unitarily equivalent to the closure of $(\mathcal{L} + V_\alpha, C_c^\infty)$.*

Proof. If $\alpha \geq 2$, then $\nabla_{\mathcal{H}} w_\alpha \in L_{\text{loc}}^\infty$ and $\mathcal{L} w_\alpha \in L_{\text{loc}}^Q \subset L_{\text{loc}}^2$ by (2.7) and (2.8), while V_α is bounded from below by Lemma 4.5. The conclusion follows from Proposition 3.4. $\square$

Remark 4.8. When $0 < \alpha < 2$, V_α is not bounded from below, and we do not know whether $(\mathcal{L} + V_\alpha, C_c^\infty)$ is essentially self-adjoint. Therefore we cannot conclude that $\mathcal{L}^{w_\alpha}$ is unitarily equivalent to the closure of $(\mathcal{L} + V_\alpha, C_c^\infty)$. However, since the weight w_α is smooth in G^* , the operators $(\mathcal{L}^{w_\alpha}, C_c^\infty(G^*))$ and $(\mathcal{L} + V_\alpha, C_c^\infty(G^*))$ are unitarily equivalent. As we shall see this will suffice to show that no self-adjoint extension of $(\mathcal{L}^{w_\alpha}, C_c^\infty(G^*))$ has discrete spectrum when $0 < \alpha \leq 2$.

We answer yes throughout the open range. The negative part of the potential is compactly supported and strictly milder than the homogeneous inverse-square scale. A quantitative resolvent estimate therefore turns the Kato inequality already proved in the source into a strict contraction of the deficiency space.

2 A general criterion

Let G be a stratified group of homogeneous dimension $Q \geq 4$, let $\mathcal{L} = -\sum_j X_j^2$ be its positive sub-Laplacian, and let Haar measure be fixed. We use the self-adjoint closure of $(\mathcal{L}, C_c^\infty(G))$.

Theorem 1. *Let $V: G \rightarrow \mathbb{R}$ belong to $L_{\text{loc}}^2(G)$. If its negative part $W = V_- := \max\{-V, 0\}$ belongs to $L^p(G)$ for some $p > Q/2$, then*

$$(\mathcal{L} + V, C_c^\infty(G))$$

is essentially self-adjoint on $L^2(G)$.

The global L^p hypothesis is convenient rather than optimal. In the application below W is compactly supported.

3 Proof intuition

The source proves essential self-adjointness when V is bounded below by applying Kato's inequality to a deficiency vector. Here V tends to $-\infty$ at the identity, so that argument cannot discard the term $V_-|f|$. Instead, apply the free resolvent $R_\lambda = (\mathcal{L} + \lambda)^{-1}$. Heat-kernel scaling gives

$$\|WR_\lambda\|_{2 \rightarrow 2} \leq C\|W\|_p \lambda^{-1+Q/(2p)} = o(1).$$

Because $p > Q/2$, this norm is strictly less than one for large λ . Kato's inequality says that a deficiency vector satisfies $(\mathcal{L} + \lambda)|f| \leq W|f|$. Testing against $R_\lambda \psi$ then yields $\|f\|_2 \leq \|WR_\lambda\|_{2 \rightarrow 2} \|f\|_2$, forcing $f = 0$.

4 Resolvent estimate

Lemma 2. *If $p > Q/2$ and $W \in L^p(G)$, then, for $\lambda > 0$,*

$$\|W(\mathcal{L} + \lambda)^{-1}\|_{2 \rightarrow 2} \leq C_{G,p} \|W\|_p \lambda^{-1+Q/(2p)}. \quad (1)$$

Proof. Put $r = 2p/(p-2)$, so that $1/2 = 1/p + 1/r$. The sub-Laplacian heat kernel has the homogeneous scaling

$$h_t(x) = t^{-Q/2} h_1(\delta_{t^{-1/2}} x).$$

Young's convolution inequality consequently gives

$$\|e^{-t\mathcal{L}}\|_{2 \rightarrow r} \leq C t^{-\frac{Q}{2}(\frac{1}{2} - \frac{1}{r})} = C t^{-Q/(2p)}.$$

Using the semigroup representation of the resolvent,

$$\|(\mathcal{L} + \lambda)^{-1}\|_{2 \rightarrow r} \leq C \int_0^\infty e^{-\lambda t} t^{-Q/(2p)} dt = C_{G,p} \lambda^{-1+Q/(2p)}.$$

Hölder's inequality for multiplication by W proves (1). The integral is finite precisely because $p > Q/2$. $\square$

5 Proof of the criterion

First, the first-order homogeneous Sobolev inequality and interpolation imply, for $s = 2p/(p-1) < 2Q/(Q-2)$ and every $\eta > 0$,

$$\|u\|_s^2 \leq \eta \|\mathcal{L}^{1/2} u\|_2^2 + C_\eta \|u\|_2^2, \quad u \in C_c^\infty(G).$$

Thus

$$\int_G W|u|^2 \leq \|W\|_p \|u\|_s^2 \leq \eta \|W\|_p \|\mathcal{L}^{1/2} u\|_2^2 + C_\eta \|W\|_p \|u\|_2^2.$$

It follows that the symmetric operator $T = (\mathcal{L} + V, C_c^\infty(G))$ is bounded from below.

Choose $\lambda > 0$ so large that $-\lambda$ lies below this lower bound and also

$$c_\lambda := \|W(\mathcal{L} + \lambda)^{-1}\|_{2 \rightarrow 2} < 1. \quad (2)$$

It is enough, by the semibounded deficiency criterion, to show that $\ker(T^* + \lambda) = \{0\}$.

Take f in this kernel. Distributionally,

$$\mathcal{L}f + Vf + \lambda f = 0.$$

Since $V \in L^2_{\text{loc}}$ and $f \in L^2$, both Vf and $\mathcal{L}f$ belong to L^1_{loc} . The sub-Laplacian Kato inequality proved in [1, Proposition 1.3] gives

$$(\mathcal{L} + \lambda)|f| \leq W|f| \tag{3}$$

in the sense of distributions.

Let $0 \leq \psi \in C_c^\infty(G)$ and put $g = (\mathcal{L} + \lambda)^{-1}\psi$. Positivity of the heat semigroup gives $g \geq 0$; moreover $g \in \text{Dom}(\mathcal{L})$, $\nabla_{\mathcal{H}}g \in L^2$, and hypoellipticity makes g smooth. To justify testing (3) against g , let χ_R be the standard nonnegative exhausting cutoffs with

$$\chi_R \rightarrow 1, \quad \|\nabla_{\mathcal{H}}\chi_R\|_\infty \leq C/R, \quad \|\mathcal{L}\chi_R\|_\infty \leq C/R^2.$$

Test against $\chi_R g$ and expand

$$(\mathcal{L} + \lambda)(\chi_R g) = \chi_R \psi + (\mathcal{L}\chi_R)g - 2\nabla_{\mathcal{H}}\chi_R \cdot \nabla_{\mathcal{H}}g.$$

The last two terms vanish after pairing with $|f|$ as $R \rightarrow \infty$, by Cauchy–Schwarz and the displayed cutoff bounds. On the right, $Wg \in L^2$ by Lemma 2, so dominated convergence is also legitimate. Hence

$$\langle |f|, \psi \rangle \leq \langle |f|, W(\mathcal{L} + \lambda)^{-1}\psi \rangle \leq c_\lambda \|f\|_2 \|\psi\|_2.$$

Taking the supremum over nonnegative unit vectors in the dense cone $C_c^\infty(G)_+$ gives

$$\|f\|_2 \leq c_\lambda \|f\|_2.$$

By (2), $f = 0$. This proves Theorem 1.

6 Application to the source potential

Let G now be a Métivier group, written as $\mathbb{R}^{2n} \times \mathbb{R}^m$, with homogeneous dimension $Q = 2n + 2m \geq 4$, and let

$$N(x, t) = (|x|^4 + 16|t|^2)^{1/4}.$$

For $w_\alpha = e^{-N^\alpha}$, let V_α be the potential in [1].

Corollary 3. *For every $0 < \alpha < 2$, the operator*

$$(\mathcal{L} + V_\alpha, C_c^\infty(G))$$

is essentially self-adjoint. Consequently the weighted form operator $\mathcal{L}^{w_\alpha}$ is unitarily equivalent, under $U_\alpha f = f\sqrt{w_\alpha}$, to the closure of this Schrödinger operator. Thus Remark 4.8 has an affirmative answer in its full stated range.

Proof. Lemma 4.5 of [1] supplies positive constants $c_{\alpha,1}$ and $c_{\alpha,2}$ such that, off the identity,

$$V_\alpha(x, t) \geq N^{2\alpha-4}|x|^2 \left(c_{\alpha,1} - \frac{c_{\alpha,2}}{N^\alpha} \right). \tag{4}$$

Choose R so that the bracket is nonnegative for $N \geq R$. Then $W_\alpha = (V_\alpha)_-$ is supported in $\{N \leq R\}$ and

$$0 \leq W_\alpha(x, t) \leq c_{\alpha,2} N^{\alpha-4} |x|^2 \leq C N^{\alpha-2}. \tag{5}$$

Since homogeneous balls have volume Cr^Q , (5) gives

$$W_\alpha \in L^p(G) \text{ whenever } p(2 - \alpha) < Q.$$

The interval

$$\frac{Q}{2} < p < \frac{Q}{2 - \alpha}$$

is nonempty exactly because $\alpha > 0$. Furthermore, the exact derivative bounds in the proof of source Lemma 4.5 give locally

$$|V_\alpha| \leq C(N^{2\alpha-2} + N^{\alpha-2}).$$

This belongs to L^2_{loc} because $2(2 - \alpha) < Q$. Theorem 1 proves essential self-adjointness.

It remains only to record the unitary statement, since the weight is not smooth at the identity. Put $h = \sqrt{w_\alpha}$. For $\phi \in C_c^\infty(G)$,

$$h\nabla_{\mathcal{H}}(h^{-1}\phi) = \nabla_{\mathcal{H}}\phi + \frac{\alpha}{2}\phi N^{\alpha-1}\nabla_{\mathcal{H}}N \in L^2.$$

The last inclusion follows from $Q \geq 4$ and $\alpha > 0$. The distributional product rule, first off the identity and then across it by local integrability, yields

$$\mathcal{L}_1^{w_\alpha}(h^{-1}\phi) = h^{-1}(\mathcal{L} + V_\alpha)\phi \in L^2(w_\alpha).$$

The domain characterization in [1, Theorem 2.3] therefore shows that $h^{-1}\phi \in \text{Dom}(\mathcal{L}^{w_\alpha})$. Hence $U_\alpha \mathcal{L}^{w_\alpha} U_\alpha^{-1}$ is a self-adjoint extension of $(\mathcal{L} + V_\alpha, C_c^\infty)$. Essential self-adjointness forces it to equal the closure, completing the proof. $\square$

7 Verification and novelty

The exponent-window script checks the strict inequalities, radial integrability exponent, resolvent decay exponent, and local L^2 condition for the critical dimension $Q = 4$ and several higher dimensions. It is a sanity check; the proof above is analytic.

A bounded search through 13 August 2026 covered the run indexes, the exact open phrase, arXiv:1610.09850, the paper title, Métivier Schrödinger potentials, and essential self-adjointness. It found the source and later papers citing it, but no explicit answer to Remark 4.8. This is not an exhaustive priority search.

Human-review recommendation. Check first the cutoff passage from (3) to the resolvent test, and then the distributional conjugation across the identity in the final paragraph. All exponent inequalities are strict for every $\alpha > 0$.

References

- [1] T. Bruno and M. Calzi, *Weighted sub-Laplacians on Métivier groups: essential self-adjointness and spectrum*, Proc. Amer. Math. Soc. 145 (2017), 3579–3594; arXiv:1610.09850.
- [2] G. B. Folland, *Subelliptic estimates and function spaces on nilpotent Lie groups*, Ark. Mat. 13 (1975), 161–207.
- [3] M. Reed and B. Simon, *Methods of Modern Mathematical Physics II: Fourier Analysis, Self-Adjointness*, Academic Press, 1975.